Chapter 14

Introduction to Mechanism Design

Mechanism design can be viewed as the *reverse engineering* of games or equivalently as the *art of designing the rules of a game to achieve a specific desired outcome*. The main focus of mechanism design is to create institutions or protocols that satisfy certain desired objectives, assuming that the individual agents, interacting through the institution, will act strategically and may hold private information that is relevant to the decision at hand. We commence our discussion of mechanism design with this chapter. We introduce mechanisms through simple examples and trace the evolution of mechanism design theory from the 1960s. Mechanisms induce a game among strategic agents in order to realize a system-wide objective or social choice function, in an equilibrium of the game. Mechanisms could be direct or indirect. In this chapter, we define the key notions of *social choice function*, *direct mechanism*, and *indirect mechanism*. We provide several illustrative examples of social choice functions.

14.1 Mechanism Design: Common Examples and History

Mechanism design is concerned with settings where a policy maker (or social planner) faces the problem of aggregating the *announced preferences* of multiple agents into a collective (or social), system-wide decision when the *actual preferences* of the agents are not publicly known. Thus mechanism design requires the designer to solve a *decision or optimization problem with incomplete information or incomplete specification*. The essential technique that mechanism design uses is to induce a game among the agents in such a way that in an equilibrium of the induced game, the desired system-wide solution is implemented. Informally, a mechanism makes the players do what the social planner would like them to do.

Mechanisms: Two Common Examples

Mechanisms have been used and practiced from times immemorial. For example, auctions, which provide a popular instance of mechanisms, have been in vogue for several centuries.

Cake Cutting Problem

The following popular stories capture the idea behind mechanisms quite strikingly (see Figure 14.1). The first story is that of a mother with two kids, who has to design a mechanism to make her kids share a cake equally. The mother is the social planner in this case. If the mother slices the cake into two equal pieces and distributes one piece to each of the kids, the solution is not necessarily acceptable to the kids because each kid will be left with the perception that he/she got the smaller of the two pieces. On the other hand, consider the following mechanism: (1) One of the kids would slice the cake into two pieces and (2) the other kid gets the chance to pick up any of the pieces, leaving the remaining piece to the kid who sliced the cake into two pieces. Child 1, who cuts the cake will slice it exactly into two equal halves (in *his* eyes), as any other division will leave him with the smaller piece (since child 2 will pick the larger of the two slices). Child 2 is happy because she gets to choose and also chooses what in *her* eyes is the larger of the two slices. This mechanism implements the desirable outcome of the kids sharing the cake equally and further each kid has every reason to be happy about this mechanism.

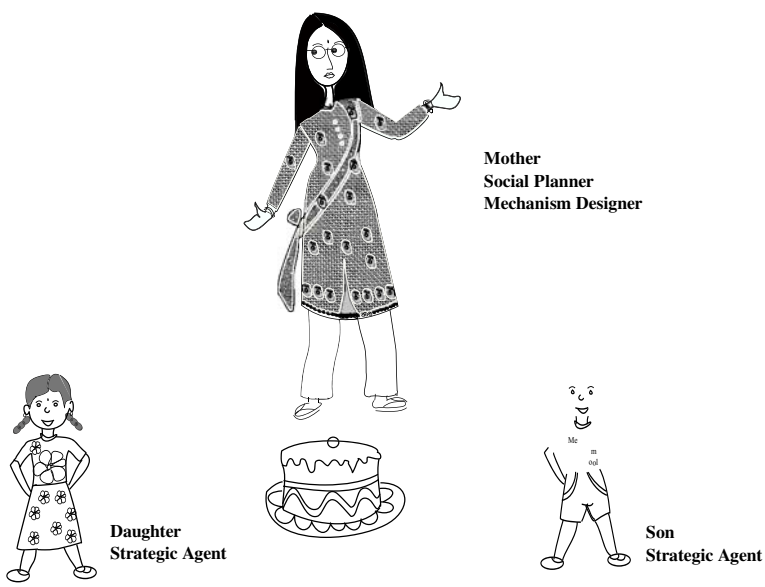

Fig. 14.1: Mechanism design for cake cutting

Baby's Mother Problem

The second story is from ancient wisdom. This is attributed to several wise people, notably King Solomon. In India, it is attributed independently to (a) Birbal who

was an adviser to King Akbar in the late 1500s and to (b) Tenali Rama, who was a popular poet and adviser in the court of King Sri Krishna Devaraya of the Vijayanagara dynasty in the early 1500s. In this fable, two women come to the king with a baby, each claiming to be the baby's mother, seeking justice (see Figure 14.2). The clueless king turns to his adviser to solve the problem. The adviser suggests the following mechanism: slice the baby into two equal pieces and give one piece each to the two mothers. Upon which, one of the women (the real mother) immediately pleads with the king not to slice the baby. The adviser and the king, after watching the reactions and body languages of the two mothers immediately ordered that the baby be handed over to the real mother. This is an example of a truth elicitation mechanism.

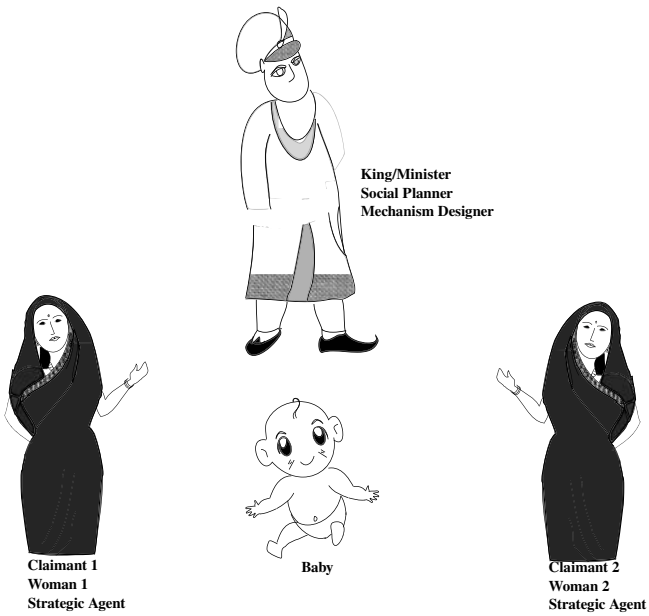

Fig. 14.2: Mechanism design for truth elicitation

Mechanisms such as above are ubiquitous in everyday life. We implicitly or explicitly use mechanisms in many activities in day-to-day life, be it family related, profession related, or society related. The emergence of game theory during the 1940s and 1950s helped develop a formal theory of mechanism design starting from the 1960s.

Mechanism Design: A Brief History

Leonid Hurwicz (Nobel laureate in Economic Sciences in 2007) first introduced the notion of mechanisms with his work in 1960 [1]. He defined a mechanism as a communication system in which participants send messages to each other and perhaps to

a *message center* and a pre-specified rule assigns an outcome (such as allocation of goods and payments to be made) for every collection of received messages. William Vickrey (Nobel laureate in Economic Sciences in 1996) wrote a classic paper in 1961 [2] which introduced the celebrated Vickrey auction (second price auction). To this day, the Vickrey auction continues to enjoy a special place in the annals of mechanism design and can be described as one of the earliest landmarks in mechanism design. John Harsanyi (Nobel laureate in Economic Sciences in 1994 jointly with John Nash and Reinhard Selten) developed the theory of games with incomplete information, in particular Bayesian games, through a series of three seminal papers in 1967-68 [3, 4, 5]. Harsanyi's work later proved to be of foundational value to mechanism design. Hurwicz [6] introduced the key notion of incentive compatibility in 1972. This notion allowed mechanism design to incorporate the incentives of rational players and opened up mechanism design. Edward Clarke [7] and Theodore Groves [8] came up with a generalization of Vickrey mechanisms and helped define a broad class of so called dominant strategy incentive compatible mechanisms in the quasi-linear environment.

Leonid Hurwicz, Eric Maskin, and Roger Myerson were jointly awarded the Nobel prize in Economic Sciences in 2007 for having laid the foundations of mechanism design theory. Hurwicz, born in 1917, is the oldest winner of the Nobel prize. It was Hurwicz who first introduced the notion of mechanisms with his work in 1960 [1]. He defined a mechanism as a communication system in which participants send messages to each other and perhaps to a *message center* and a pre-specified rule assigns an outcome (such as allocation of goods and payments to be made) for every collection of received messages.

Hurwicz introduced the key notion of incentive compatibility in 1972 [6]. This notion allowed mechanism design to incorporate the incentives of rational players and opened up the area of mechanism design. The notion of incentive compatibility plays a central role in the revelation theorem, which is a fundamental result in mechanism design theory. Hurwicz is also credited with many important possibility and impossibility results in mechanism design. For example, he showed that, in a standard exchange economy, no incentive compatible mechanism that satisfies individual rationality can produce Pareto optimal outcomes. Hurwicz's work in game theory and mechanism design demonstrated, beyond doubt, the value of using analytical methods in modeling economic institutions.

Hurwicz was, until his demise on June 24, 2008, Regents Professor Emeritus in the Department of Economics at the University of Minnesota. He taught there in the areas of welfare economics, public economics, mechanisms and institutions, and mathematical economics.

Eric Maskin is credited with many pioneering contributions to mechanism design. One of his most creative contributions was his work on implementation theory, which addresses the following problem: Given a social goal, can we characterize when we can design a mechanism whose equilibrium outcomes coincide with the outcomes that are desirable according to that goal? Maskin [9] gave a general solution to this problem. He brilliantly showed that if social goals are to be implementable, then they must satisfy a certain kind of *monotonicity* (Maskin Monotonicity). He also showed that monotonicity guarantees implementation under certain mild conditions (at least three players and no veto power). He has also made major contributions to dynamic games. One of his early contributions was to formalize the Revelation Theorem to the setting of Bayesian incentive compatible mechanisms.

Maskin was born on December 12, 1950, in New York City. He earned an A.B. in Mathematics and a Ph.D. in Applied Mathematics from Harvard University in 1976. He taught at the Massachusetts Institute of Technology during 1977–1984 and at Harvard University during 1985–2000. During 2000-2011, he was the Albert O. Hirschman Professor of Social Science at the Institute for Advanced Study in Princeton, NJ, USA. Since 2011, he is Adams University Professor in the Department of Economics at the Harvard University, Cambridge, Massachusetts, USA.

Roger Bruce Myerson has contributed to several areas in game theory and mechanism design, and his contributions have left a deep impact in the area. He was instrumental in conceptualizing and proving the revelation theorem in mechanism design for Bayesian implementations in its greatest generality. His work on optimal auctions in 1981 is a landmark result and has led to an extensive body of further work in the area of optimal auctions. He has also made major contributions in bargaining with incomplete information and cooperative games with incomplete information. His textbook *Game Theory: Analysis of Conflict* is a scholarly and comprehensive reference text that embodies all important results in game theory in a rigorous, yet insightful way. Myerson has also worked on economic analysis of political institutions and written several influential papers in this area including recently on democratization and the Iraq war.

Myerson was born on March 29, 1951. He received his A.B., S.M., and Ph.D., all in Applied Mathematics from Harvard University. He completed his Ph.D. in 1976, working with the legendary Kenneth Arrow. He was a Professor of Economics at the Kellogg School of Management in Northwestern University during 1976-2001. Since 2001, he has been the Glen A. Lloyd Distinguished Service Professor of Economics at the University of Chicago.

There were two major advances in mechanism design in the 1970s. The first was the *revelation principle* which essentially showed that direct mechanisms are the same as indirect mechanisms. This meant that mechanism theorists needed to worry only about direct mechanisms, leaving the development of real-world mechanisms (which are mostly indirect mechanisms) to mechanism designers and practitioners. Alan Gibbard [10] formulated the revelation principle for dominant strategy incentive compatible mechanisms. This was later extended to Bayesian incentive compatible mechanisms through several independent efforts [11] – Eric Maskin and Roger Myerson (both Nobel laureates in Economic Sciences in 2007) had a leading role to play in this. In fact, Myerson developed the revelation principle in its greatest generality [11]. The second major advance in mechanism design in the 1970s was on *implementation theory* which addresses the following problem: can a mechanism be designed so that all its equilibria are optimal? Maskin [9] gave the first general solution to this problem.

Mechanism design has made phenomenal advances during 1980s, 1990s, 2000s, and during the past few years. It has found widespread applicability in a variety of disciplines. These include: design of auctions, markets, and trading institutions [11, 12, 13, 14], regulation and auditing [11], social choice theory [11], computer science [15], resource allocation to strategic agents, electronic commerce and web-based applications [16], etc. This list is by no means exhaustive.

14.2 Mechanism Design Environment

The following provides a general setting for formulating, analyzing, and solving mechanism design problems.

- There are n agents, $1, 2, \ldots, n$, with $N = \{1, 2, \ldots, n\}$. The agents are rational and intelligent, and interact strategically among themselves towards making a collective decision.
- X is a set of *alternatives* or *outcomes*. The agents are required to make a collective choice from the set X.
- Prior to making the collective choice, each agent privately observes his preferences over the alternatives in X. This is modeled by supposing that agent i privately observes a parameter or signal θ_i that determines his preferences. The value of θ_i is known to agent i and may not be known to the other agents. θ_i is called a *private value* or *type* of agent i.
- We denote by Θ_i the set of private values of agent i, $i = 1, 2, \ldots, n$. The set of all type profiles is given by $\Theta = \Theta_1 \times \ldots \times \Theta_n$. A typical type profile is represented as $\theta = (\theta_1, \ldots, \theta_n)$.
- We assume that there is a common prior $\mathbb{P} \in \Delta(\Theta)$. To ensure consistency of beliefs, individual belief functions $p_i : \Theta_i \to \Delta(\Theta_{-i})$ (where Θ_{-i} is the set of type profiles of agents other than i) can all be derived from the common prior.

- Individual agents have preferences over outcomes that are represented by a utility function $u_i : X \times \Theta_i \to \mathbb{R}$. Given $x \in X$ and $\theta_i \in \Theta_i$, the value $u_i(x, \theta_i)$ denotes the payoff that agent i, having type $\theta_i \in \Theta_i$, receives from an outcome $x \in X$. In the more general case, u_i depends not only on the outcome and the type of player i, but could depend on the types of the other players as well, and so $u_i : X \times \Theta \to \mathbb{R}$. We restrict our attention to the former case in this book since most situations discussed in this book fall into the former category.
- The set of outcomes X, the set of players N, the type sets Θ_i $(i = 1, \ldots, n)$, the common prior distribution $\mathbb{P} \in \Delta(\Theta)$, and the payoff functions u_i $(i = 1, \ldots, n)$ are assumed to be *common knowledge* among all the players. The specific type θ_i observed by agent i is private information of agent i.

Social Choice Functions

Since the preferences of the agents depend on the realization of their types $\theta = (\theta_1, \ldots, \theta_n)$, it is logical and natural to make the collective decision depend on θ. This leads to the definition of a social choice function.

Definition 14.1 (Social Choice Function). *Suppose $N = \{1, 2, \ldots, n\}$ is a set of agents with the type sets $\Theta_1, \Theta_2, \ldots, \Theta_n$ respectively. Given a set of outcomes X, a social choice function is a mapping $f : \Theta_1 \times \ldots \times \Theta_n \to X$ that assigns to each possible type profile $(\theta_1, \theta_2, \ldots, \theta_n)$, an outcome from the set X. The outcome corresponding to a type profile is called a social choice or collective choice for that type profile.*

We will be presenting several examples of social choice functions subsequently in this chapter. We provide an immediate example here in an informal way.

Example 14.1 (Shortest Path Problem with Incomplete Information). Consider a connected directed graph with a source vertex and a destination vertex identified. Let the graph have n edges, each owned by a rational and intelligent agent. Let the set of agents be denoted by $N = \{1, 2, \ldots, n\}$. Assume that the cost of the edge is private information of the agent owning the edge and let θ_i be this private information for agent i $(i = 1, 2, \ldots, n)$. Let us say that a social planner is interested in finding a shortest path from the source vertex to the destination vertex. The social choice function here maps each type profile $(\theta_1, \ldots, \theta_n)$ (which is in fact a profile of edge costs) to a shortest path corresponding to those edge costs. The social planner knows everything about the graph except the costs of the edges. So, the social planner first needs to extract this information from each agent and then find a shortest path from the source vertex to the destination vertex. Thus there are two problems facing the social planner, which are described below. □

Preference Elicitation Problem

Consider a social choice function $f : \Theta_1 \times \ldots \times \Theta_n \to X$. The types $\theta_1, \ldots, \theta_n$ of the individual agents are private information of the agents. Hence for the social choice $f(\theta_1, \ldots, \theta_n)$ to be chosen when the individual types are $\theta_1, \ldots, \theta_n$, each agent must disclose its true type to the social planner. However, given a social choice function f, a given agent may not find it in its best interest to reveal this information truthfully. This is called the *preference elicitation* problem or the *information revelation* problem.

Preference Aggregation Problem

Once all the agents report their types, the profile of reported types has to be transformed to an outcome, based on the social choice function. Let θ_i be the true type and $\hat{\theta}_i$ the reported type of agent i ($i = 1, \ldots, n$). The process of computing $f(\hat{\theta}_1, \ldots, \hat{\theta}_n)$ is called the *preference aggregation* problem.

Example 14.2. In the current example of shortest path problem with incomplete information, the preference elicitation problem is to elicit the true values of the costs of the edges from the respective edge owners. The preference aggregation problem is to compute a shortest path from the source vertex to the destination vertex, given the structure of the graph and the (reported) costs of the edges. The preference aggregation problem is often an optimization problem like for instance in the current example. □

Figure 14.3 provides a pictorial representation of all the elements in a mechanism design environment.

14.3 Direct and Indirect Mechanisms

One can view mechanism design as the process of solving an incompletely specified optimization problem where the specification is first elicited and then the underlying optimization problem or decision problem is solved. Specification elicitation is basically the preference elicitation or type elicitation problem. To elicit the type information from the agents in a truthful way, there are broadly two kinds of approaches, which are aptly called *direct mechanisms* and *indirect mechanisms*. We define these below. In these definitions, we assume that the set of agents N, the set of outcomes X, the sets of types $\Theta_1, \ldots, \Theta_n$, the common prior $\mathbb{P} \in \Delta(\Theta)$, and the utility functions $u_i : X \times \Theta_i \to \mathbb{R}$ are given and are common knowledge.

Definition 14.2 (Direct Mechanism). *Suppose $f : \Theta_1 \times \ldots \times \Theta_n \to X$ is a social choice function. A direct mechanism (also called a direct revelation mechanism) corresponding to f consists of the tuple $(\Theta_1, \Theta_2, \ldots, \Theta_n, f(.))$.*

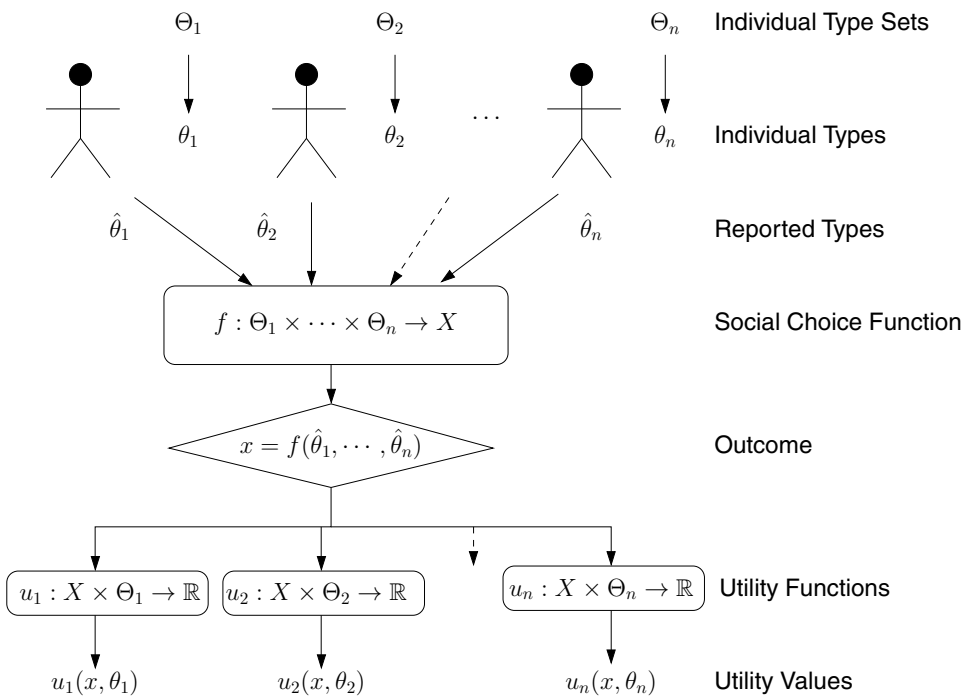

Fig. 14.3: Mechanism design environment

The idea of a direct mechanism is to *directly* seek the type information from the agents by asking them to reveal their true types.

Definition 14.3 (Indirect Mechanism). *An indirect mechanism (also called an indirect revelation mechanism) consists of a tuple* $(S_1, S_2, \ldots, S_n, g(.))$ *where* S_i *is a set of possible actions for agent* i $(i = 1, 2, \ldots, n)$ *and* $g : S_1 \times S_2 \times \ldots \times S_n \to X$ *is a function that maps each action profile to an outcome.*

The idea of an indirect mechanism is to provide a choice of actions to each agent and specify an outcome for each action profile. This induces a game among the players and the strategies played by the agents in an equilibrium of this game will indirectly reflect their original types. The idea behind and the difference between direct mechanisms and indirect mechanisms will become clear in the next chapter when we discuss implementation of social choice functions by mechanisms.

Note that a direct mechanism corresponding to $f : \Theta_1 \times \ldots \times \Theta_n \to X$ is a special case of an indirect mechanism $(S_1, S_2, \ldots, S_n, g(.))$ with $S_i = \Theta_i \; \forall i \in N$ and $g = f$.

14.4 Examples of Social Choice Functions

In this section, we provide several examples to illustrate the notion of a social choice function.

Example 14.3 (Supplier Selection). Suppose there is a buyer who wishes to procure a certain volume of an item that is produced by two suppliers, call them 1 and 2. We have $N = \{1, 2\}$. Supplier 1 is known to use technology a_1 to produce these items, while supplier 2 uses one of two possible technologies, (1) a superior, high end technology a_2 and (2) a low end technology b_2. The technology elements could be taken as the types of the suppliers, so we have $\Theta_1 = \{a_1\}$; $\Theta_2 = \{a_2, b_2\}$. See Figure 14.4.

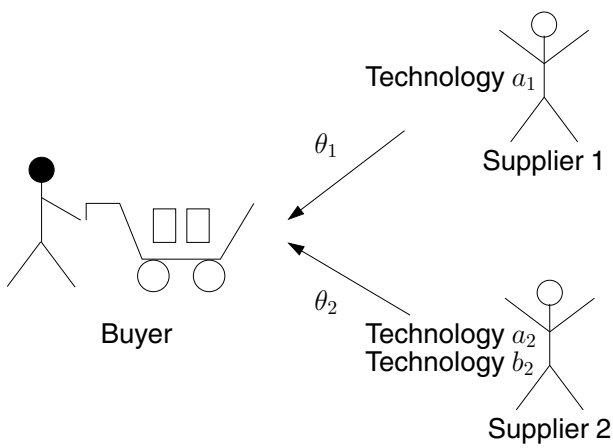

Fig. 14.4: A procurement scenario with two suppliers

Let us define three outcomes (alternatives) x, y, z for this situation. The alternative x means that the entire volume required is sourced from supplier 1 while the alternative z means that the entire volume required is sourced from supplier 2. The alternative y indicates that 50% of the requirement is sourced from supplier 1 and the rest is sourced from supplier 2.

Assume that the buyer already has a long-standing relationship with supplier 1, this supplier is the preferred one. However, because of the superiority of technology a_2 over a_1 and b_2, the buyer may like to source at least a certain fraction of requirement from supplier 2 if supplier 2 uses technology a_2. The payoff functions below reflect the above facts:

$$u_1(x, a_1) = 100; \;\; u_1(y, a_1) = 50; \;\; u_1(z, a_1) = 0$$

$$u_2(x, a_2) = 0; \;\; u_2(y, a_2) = 50; \;\; u_2(z, a_2) = 100$$

$$u_2(x, b_2) = 0; \;\; u_2(y, b_2) = 50; \;\; u_2(z, b_2) = 25.$$

Note that $\Theta = \{(a_1, a_2), (a_1, b_2)\}$. Consider the social choice function $f(a_1, a_2) = y$ and $f(a_1, b_2) = x$. This means that when supplier 2 uses technology a_2, the buyer would like to procure from both the suppliers, whereas if supplier 2 uses technology b_2, the buyer would rather source the entire requirement from supplier 1. Note that the technology used by supplier 2 is private information and is not known to the buyer.

Likewise, there are eight other social choice functions that one can define here. These are included below.

$$\begin{aligned}
f(a_1, a_2) = x; \quad & f(a_1, b_2) = x \\
f(a_1, a_2) = x; \quad & f(a_1, b_2) = y \\
f(a_1, a_2) = x; \quad & f(a_1, b_2) = z \\
f(a_1, a_2) = y; \quad & f(a_1, b_2) = y \\
f(a_1, a_2) = y; \quad & f(a_1, b_2) = z \\
f(a_1, a_2) = z; \quad & f(a_1, b_2) = x \\
f(a_1, a_2) = z; \quad & f(a_1, b_2) = y \\
f(a_1, a_2) = z; \quad & f(a_1, b_2) = z.
\end{aligned}$$

It would be interesting to look into the practical meaning and implications of these social choice functions. □

Example 14.4 (Selling an Indivisible Object). A typical selling situation involves a seller, a pool of buyers, and objects to be sold. We consider a simple abstraction of the problem by considering a selling agent (call the agent 0) and two buying agents (call them 1 and 2), so we have $N = \{0, 1, 2\}$. See Figure 14.5.

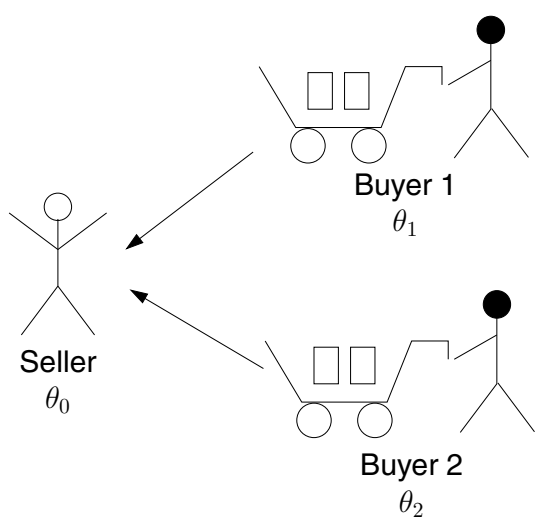

Fig. 14.5: A selling scenario with two buying agents

The seller wishes to sell an indivisible object to one of the buyers in return for a monetary payment. An outcome here can be represented by $x = (y_0, y_1, y_2, t_0, t_1, t_2)$ where y_0, y_1, y_2 indicate the allocations and t_0, t_1, t_2 indicate the payments. For $i = 0, 1, 2$, y_i takes value 1 if the object is allocated to player i and takes the value 0 otherwise; t_i is the payment

received by agent i. The set X of all feasible outcomes is given by

$$X = \{(y_0, y_1, y_2, t_0, t_1, t_2) : y_i \in \{0,1\}, \sum_{i=0}^{2} y_i = 1, t_i \in \mathbb{R},\ i \in N\}.$$

For $x = (y_0, y_1, y_2, t_0, t_1, t_2)$, we can define the utilities for buyers $1, 2$ in a natural way as follows:

$$u_i(x, \theta_i) = u_i((y_0, y_1, y_2, t_0, t_1, t_2), \theta_i) = y_i\theta_i + t_i\ ;\ \ i = 1, 2$$

where $\theta_i \in \mathbb{R}_+$ can be viewed as agent i's valuation of the object. Such utility functions are said to be of *quasi-linear* form (because it is linear in some of the variables and possibly non-linear in the other variables). We will be studying such utility forms quite extensively from Chapter 18 onwards. We make the following assumptions regarding valuations.

- The seller has a *known* value θ_0 for the object. That is, $\Theta_0 = \{\theta_0\}$.
- Let $\Theta_i \subseteq \mathbb{R}_+$ be the set of all possible valuations of buyer i ($i = 1, 2$) for the object. The type sets are common knowledge. The exact type of a buyer before playing the game is private information of the buyer. The type of a buyer is to be viewed as the *willingness to pay* (maximum price above which the buyer is not interested in buying the item).

Consider the following social choice function.

- The selling agent allocates the object to the buyer with the highest willingness to buy. If both the buyers have the same type, the seller will allocate the object to buyer 1.
- The allocated buyer, say buyer i, pays an amount equal to θ_i to the selling agent.

The above social choice function $f(\theta) = (y_0(\theta), y_1(\theta), y_2(\theta), t_0(\theta), t_1(\theta), t_2(\theta))$, which we will call SCF1 from now on, can be precisely written, $\forall\theta = (\theta_0, \theta_1, \theta_2) \in \Theta_0 \times \Theta_1 \times \Theta_2$, as

$$\begin{aligned}
y_0(\theta) &= 0 \\
y_1(\theta) &= 1 \quad \text{if } \theta_1 \geq \theta_2 \\
&= 0 \quad \text{if } \theta_1 < \theta_2 \\
y_2(\theta) &= 1 \quad \text{if } \theta_1 < \theta_2 \\
&= 0 \quad \text{if } \theta_1 \geq \theta_2 \\
t_1(\theta) &= -y_1(\theta)\ \theta_1 \\
t_2(\theta) &= -y_2(\theta)\ \theta_2 \\
t_0(\theta) &= -(t_1(\theta) + t_2(\theta)).
\end{aligned}$$

Suppose we consider another social choice function, which has the same allocation rule as the one we have just studied but has a different payment rule. The winning buyer now pays the seller an amount equal to the second highest willingness to buy (as usual, the losing buyer does not pay anything to the seller). The new social choice function, which we will call SCF2, will be the following: For any $\theta = (\theta_0, \theta_1, \theta_2) \in \Theta_0 \times \Theta_1 \times \Theta_2$,

$$\begin{aligned}
y_0(\theta) &= 0 \\
y_1(\theta) &= 1 \quad \text{if } \theta_1 \geq \theta_2 \\
&= 0 \quad \text{if } \theta_1 < \theta_2 \\
y_2(\theta) &= 1 \quad \text{if } \theta_1 < \theta_2 \\
&= 0 \quad \text{if } \theta_1 \geq \theta_2 \\
t_1(\theta) &= -y_1(\theta)\ \theta_2 \\
t_2(\theta) &= -\ y_2(\theta)\ \theta_1 \\
t_0(\theta) &= -(t_1(\theta) + t_2(\theta)).
\end{aligned}$$

It is instructive to write down the above SCFs in the case of more than two buyers. □

We now define a complementary social choice function that describes the reverse of selling, namely buying. Our abstraction here consists of one buying agent and two selling agents.

Example 14.5 (Procurement of an Indivisible Object). We consider a buying agent (call the agent 0) and two selling agents (call them 1 and 2), so we have $N = \{0, 1, 2\}$. See Figure 14.6. An indivisible object is to be procured from one of the sellers in return for a monetary payment. An outcome here can be represented by $x = (y_0, y_1, y_2, t_0, t_1, t_2)$. The numbers y_0, y_1, y_2 indicate the allocations and t_0, t_1, t_2 indicate the payments. For $i = 1, 2$, y_i takes value 1 if the object is procured from seller i and takes the value 0 otherwise; y_0 takes the value 0 when the buyer receives the object and takes the value 1 if the buyer does not receive the object. t_i $(i = 0, 1, 2)$ is the payment received by agent i (or monetary transfer to agent i).

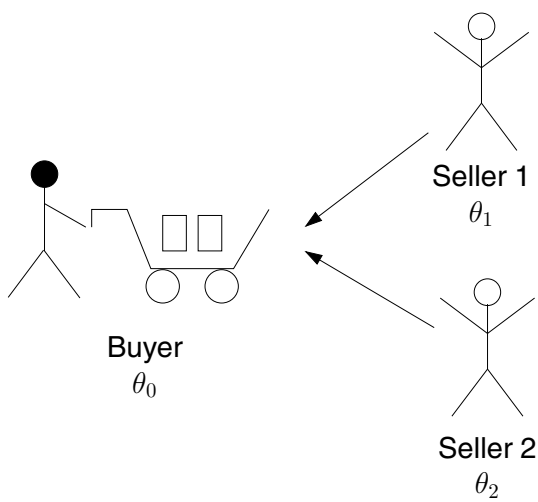

Fig. 14.6: A procurement scenario with two sellers

The set X of all feasible outcomes is given by

$$X = \{(y_0, y_1, y_2, t_0, t_1, t_2) : y_i \in \{0, 1\}, \sum_{i=0}^{2} y_i = 1, t_i \in \mathbb{R},\ i = 0, 1, 2\}.$$

For $x = (y_0, y_1, y_2, t_0, t_1, t_2)$, the utilities of the selling agents $1, 2$ can be defined in a natural way to be of the form:

$$u_i(x, \theta_i) = u_i((y_0, y_1, y_2, t_0, t_1, t_2), \theta_i) = -y_i\theta_i + t_i\ ;\ \ i = 1, 2$$

where $\theta_i \in \mathbb{R}_+$ can be viewed as seller i's valuation or cost (also called willingness to sell) of the object. We make the following assumptions regarding valuations.

- The buyer has a *known* value θ_0 for the object. That is, $\Theta_0 = \{\theta_0\}$. This valuation does not depend on the choice of the seller from whom the item is purchased.

- Let $\Theta_i \subseteq \mathbb{R}_+$ be the set of all types of agent i ($i = 1, 2$). These sets are common knowledge. However, the exact value of θ_i before playing the game is private information of player i ($i = 1, 2$. The type of a seller is to be viewed as the *willingness to sell* (minimum price below which the seller is not interested in selling the item).

Consider the following social choice function.

- The buyer buys the object from the seller with the lowest willingness to sell. If both the sellers have the same type, the buyer will buy the object from seller 1.
- The buyer pays to the allocated seller, say seller i, an amount equal to θ_i.

The above social choice function $f(\theta) = (y_0(\theta), y_1(\theta), y_2(\theta), t_0(\theta), t_1(\theta), t_2(\theta))$ can be precisely written $\forall \theta = (\theta_0, \theta_1, \theta_2) \in \Theta_0 \times \Theta_1 \times \Theta_2$ as

$$\begin{aligned}
y_0(\theta) &= 0 \\
y_1(\theta) &= 1 \quad \text{if } \theta_1 \leq \theta_2 \\
&= 0 \quad \text{if } \theta_1 > \theta_2 \\
y_2(\theta) &= 1 \quad \text{if } \theta_1 > \theta_2 \\
&= 0 \quad \text{if } \theta_1 \leq \theta_2 \\
t_1(\theta) &= y_1(\theta)\, \theta_1 \\
t_2(\theta) &= y_2(\theta)\, \theta_2 \\
t_0(\theta) &= -(t_1(\theta) + t_2(\theta)).
\end{aligned}$$

Suppose we consider another social choice function, which has the same allocation rule as the one we have just studied but has a different payment rule. The buyer now pays the winning seller a payment equal to the second lowest willingness to sell (as usual, the losing seller does not receive any payment). The new social choice function will be the following.

$$\begin{aligned}
y_0(\theta) &= 0 \\
y_1(\theta) &= 1 \quad \text{if } \theta_1 \leq \theta_2 \\
&= 0 \quad \text{if } \theta_1 > \theta_2 \\
y_2(\theta) &= 1 \quad \text{if } \theta_1 > \theta_2 \\
&= 0 \quad \text{if } \theta_1 \leq \theta_2 \\
t_1(\theta) &= y_1(\theta)\, \theta_2 \\
t_2(\theta) &= y_2(\theta)\, \theta_1 \\
t_0(\theta) &= -(t_1(\theta) + t_2(\theta)).
\end{aligned}$$

It is instructive to write down the above SCFs in the case of more than two sellers. □

Example 14.6 (Bilateral Trade). Consider two agents 1 and 2 where agent 1 is the seller of an indivisible object and agent 2 is a prospective buyer of the good. See Figure 14.7. An outcome here is of the form $x = (y_1, y_2, t_1, t_2)$ where $y_i = 1$ if agent i gets the object and t_i denotes the payment received by agent i ($i = 1, 2$). A natural set of outcomes here is

$$X = \{(y_1, y_2, t_1, t_2) : \; y_1 + y_2 = 1; \; y_1, y_2 \in \{0, 1\}\}.$$

The utility of the agent i ($i = 1, 2$) would be of the form

$$u_i((y_1, y_2, t_1, t_2), \theta_i) = y_i \theta_i + t_i.$$

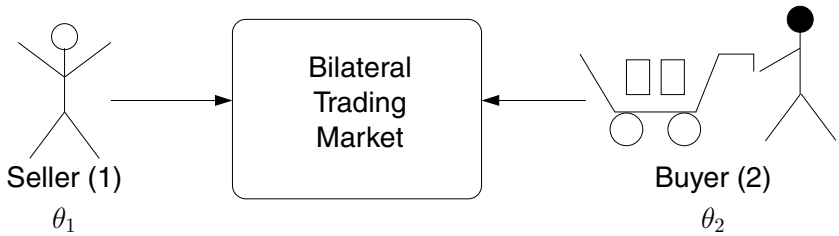

Fig. 14.7: A bilateral trade environment

The type θ_1 of agent 1 (seller) can be interpreted as the willingness to sell of the agent (minimum price at which agent 1 is willing to sell). The type θ_2 of agent 2 (buyer) has the natural interpretation of willingness to pay (maximum price the buyer is willing to pay). A social choice function here would be $f(\theta) = (y_1(\theta), y_2(\theta), t_1(\theta), t_2(\theta))$ defined as

$$\begin{aligned}
y_1(\theta_1, \theta_2) &= 1 \quad \theta_1 > \theta_2 \\
&= 0 \quad \theta_1 \leq \theta_2 \\
y_2(\theta_1, \theta_2) &= 1 \quad \theta_1 \leq \theta_2 \\
&= 0 \quad \theta_1 > \theta_2 \\
t_1(\theta_1, \theta_2) &= \quad y_2(\theta_1, \theta_2) \left(\frac{\theta_1 + \theta_2}{2} \right) \\
t_2(\theta_1, \theta_2) &= -y_2(\theta_1, \theta_2) \left(\frac{\theta_1 + \theta_2}{2} \right).
\end{aligned}$$

Note that the money the buyer pays (which is equal to the money the seller receives) in case trade happens is equal to half of $\theta_1 + \theta_2$. □

Example 14.7 (Funding a Public Project). Consider a set of individuals $N = \{1, 2, \ldots, n\}$ who have a stake in a common infrastructure, for example, a bridge, community building, Internet infrastructure, etc. For example, the agents could be firms forming a business cluster and interested in creating a shared infrastructure. The cost of the project is to be shared by the agents themselves since there is no source of external funding. Let $k = 1$ indicate that the project is taken up, with $k = 0$ indicating that the project is dropped. Let $t_i \in \mathbb{R}$ denote the payment received by agent i (which means $-t_i$ is the payment made by agent i) for each $i \in N$. Let the cost of the project be C. Since the agents have to fund the project themselves,

$$C \leq -\sum_{i \in N} t_i \quad \text{if } k = 1$$

$$0 \leq -\sum_{i \in N} t_i \quad \text{if } k = 0$$

Combining the above two possibilities, we get the condition

$$\sum_{i \in N} t_i \leq -kC \; ; \;\; k \in \{0, 1\}.$$

Thus, a natural set of outcomes for this problem is:

$$X = \{(k, t_1, \ldots, t_n) : k \in \{0, 1\}, \; t_i \in \mathbb{R} \; \forall i \in N, \; \sum_{i \in N} t_i \leq -kC\}.$$

We assume the utility of agent i, when its type is θ_i corresponding to an outcome $(k, t_1, t_2, \ldots, t_n)$ to be given by

$$u_i((k, t_1, \ldots, t_n), \theta_i) = k\theta_i + t_i.$$

The type θ_i of agent i has the natural interpretation of being the willingness to pay of agent i (maximum amount that agent i is prepared to pay) towards the project. A social choice function in this context is $f(\theta) = (k(\theta), t_1(\theta), \ldots, t_n(\theta))$ given by

$$k(\theta) = \begin{cases} 1 \text{ if } \sum\limits_{i \in N} \theta_i \geq C \\ 0 \text{ otherwise} \end{cases}$$

$$t_i(\theta) = -\left(\frac{k(\theta)C}{n}\right).$$

The way $k(\theta)$ is defined ensures that the project is taken up only if the combined willingness to pay of all the agents is at least the cost of the project. The definition of $t_i(\theta)$ above follows the egalitarian principle, namely that the agents share the cost of the project equally among themselves. □

All the examples above illustrate how a meaningful social choice choice function may be defined in a given setting.

14.5 Summary and References

In this chapter, we introduced concepts and building blocks of mechanism design with examples and definitions. In particular:

- We started with two common examples (cake cutting and baby's mother problems) and traced the history of developments in mechanism design.
- We introduced the building blocks of a mechanism design environment such as social choice functions, preference elicitation, preference aggregation, direct mechanisms, and indirect mechanisms.
- We next gave several examples to illustrate the notion of social choice functions: selecting suppliers; selling an indivisible object; buying an indivisible object; bilateral trade; and funding a public project.

The next chapter is devoted to explaining how social choice functions may be implemented by direct mechanisms and indirect mechanisms.

The examples discussed in this chapter are mainly based on the examples presented by Mas-Colell, Whinston, and Green [12] and the monograph by Narahari, Garg, Ramasuri, and Hastagiri [16]. The scientific background document available on the Nobel Prize website [11] provides an excellent introduction and overview.

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14.6 Exercises

(1) In the cake cutting problem, if there were three children rather than two, what would be some appropriate mechanisms for fair sharing of the cake?

(2) Write down the type sets, outcomes, and social choice function in the cake cutting problem.

(3) Write down the type sets, outcomes, and social choice function in the baby's mother problem.
(4) Write down the social choice functions SCF1 and SCF2 assuming one selling agent and n buying agents.
(5) Write down social choice functions analogous to SCF1 and SCF2 when we have a procurement situation with one buying agent and n selling agents.